



2025-2026

Final Project **FINTECH**

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Class : ING3 MMF



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1- DATA EXPLORATION

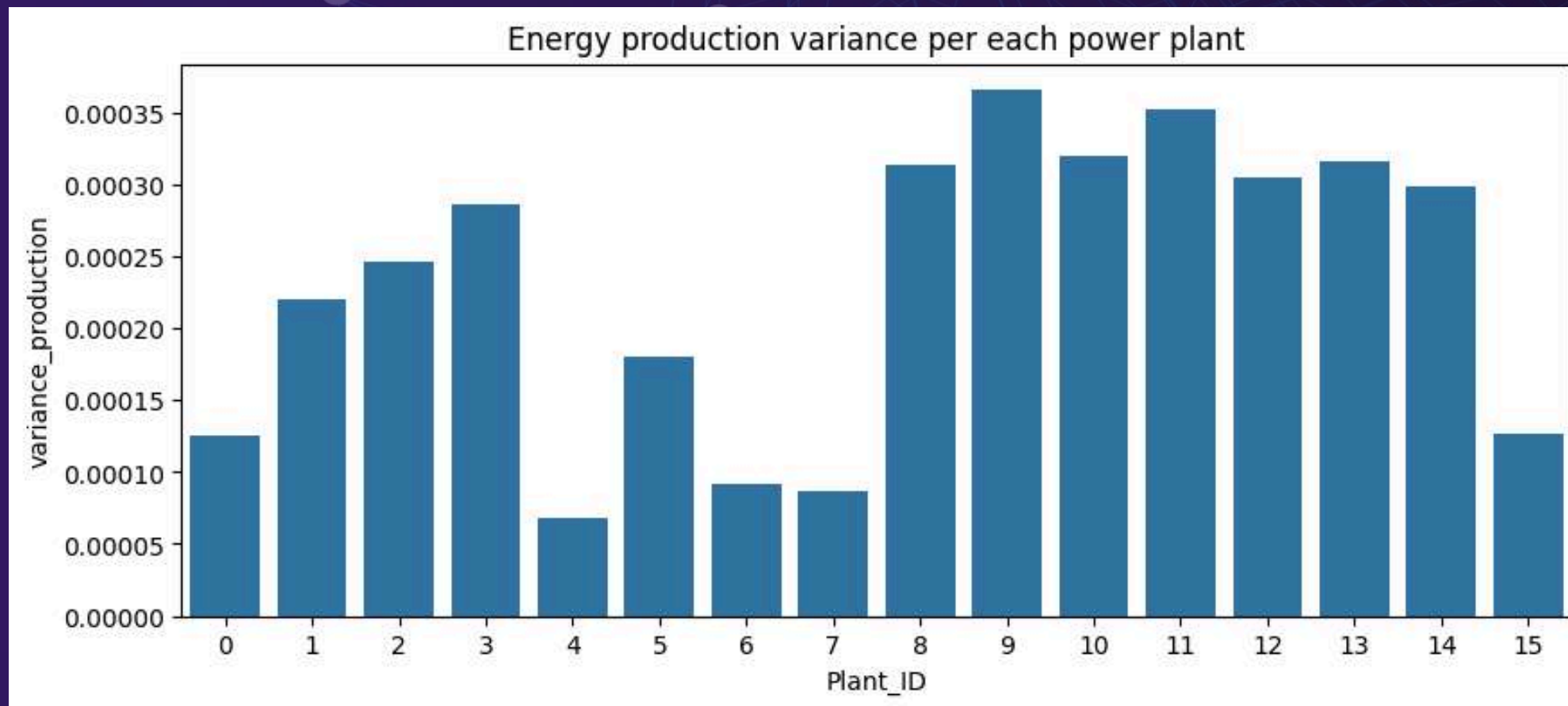
```
# Show the unique power plant  
train_set.Plant_ID.unique()  
  
... array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15])
```

count	
Plant_ID	
0	11495
1	11495
3	11495
4	11495
6	11495
5	11495
7	11495
8	11495
12	11495
9	11495
10	11495
11	11495
14	11495
13	11495
15	11495
2	4080



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PRODUCTION VARIANCE

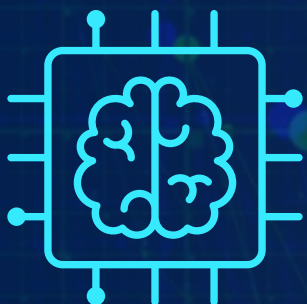


MISSING VALUES

```
*** UTC Time:0
    Plant_ID:0
    Energy_Production:0
    Benchmark:0
    Day-Night:0
    Temperature descriptor:0
    Comfort level:0
    Visibility:0
    Probability of precipitation for last hour:0
    Rainfall amount for last hour:0
    Dew point:0
    Temperature:0
    Wind direction:0
    Precipitation descriptor:0
    Sky descriptor:0
    Humidity:0
    Wind speed:0
```



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CORRELATIONS

	Visibility	Probability of precipitation for last hour	Rainfall amount for last hour	Dew point	Temperature	Humidity	Wind speed
Visibility	1.000000	-0.563152	-0.393927	-0.298848	0.095915	-0.681640	0.082980
Probability of precipitation for last hour	-0.563152	1.000000	0.622571	0.188510	0.029439	0.267744	0.271051
Rainfall amount for last hour	-0.393927	0.622571	1.000000	0.157078	0.052457	0.166451	0.206120
Dew point	-0.298848	0.188510	0.157078	1.000000	0.864787	-0.076346	0.039836
Temperature	0.095915	0.029439	0.052457	0.864787	1.000000	-0.558908	0.223713
Humidity	-0.681640	0.267744	0.166451	-0.076346	-0.558908	1.000000	-0.371277
Wind speed	0.082980	0.271051	0.206120	0.039836	0.223713	-0.371277	1.000000



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2- FEATURE ENGINEERING

Feature engineering is about transforming raw data into informative variables so that it improves the performance of models.

`"2023-06-01 10:00" → type: str`

`2023-06-01 10:00:00 → type: datetime64`

Day / Night

1 / 0

23 and 0 are far apart

23 and 0 are very close

only at time t

Using past values and short-term trends



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Then we select the variables which are exploitable, we remove redundant variables and finally we handle missing values

```
Before dropna: (176505, 296)  
After dropna: (175737, 296)
```



3- MODELE SELECTION AND TUNING

Objective: Develop a Machine Learning model to predict Energy_Production and outperform the industrial Benchmark

Validation Strategy:

- *Data Split: 70% Training / 30% Validation*
- *Chronological Split: Strictly strictly time-based (no shuffling) to respect the temporal nature of the data*
- *Metrics: RMSE (Root Mean Squared Error) and R^2 (Coefficient of Determination)*

```
Training Set : (123015, 295) (70.0%)  
Validation Set : (52722, 295) (30.0%)
```



MODELS & BENCHMARK



*Benchmark (Baseline): Current industry performance:
RMSE = 0.0071*

Candidate Algorithms:

- *Linear: Ridge, Lasso (Regularized Regression)*
- *Distance-based: K-Nearest Neighbors (KNN)*
- *Tree: Random Forest*
- *Boosting: XGBoost, LightGBM*

→ Standardization applied for linear and distance-based models



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MODEL PERFORMANCE COMPARAISON

- *Linear Models (Ridge/Lasso): Good performance ($R^2 > 0.92$), showing strong results*
- *KNN: Poorest performance among all the models ($R^2 = 0.77$)*
- *Random Forest: Robust model but with a high computational cost*
- *Boosting (XGBoost/LightGBM): Best performance with lowest error*

	Model	RMSE	R2
0	XGBoost	0.002887	0.949600
1	LightGBM	0.002931	0.948059
2	Ridge Regression	0.003061	0.943368
3	Random Forest	0.003138	0.940464
4	Lasso Regression	0.003605	0.921443
5	KNN	0.006180	0.769082
6	Benchmark (Baseline)	0.007067	0.622516



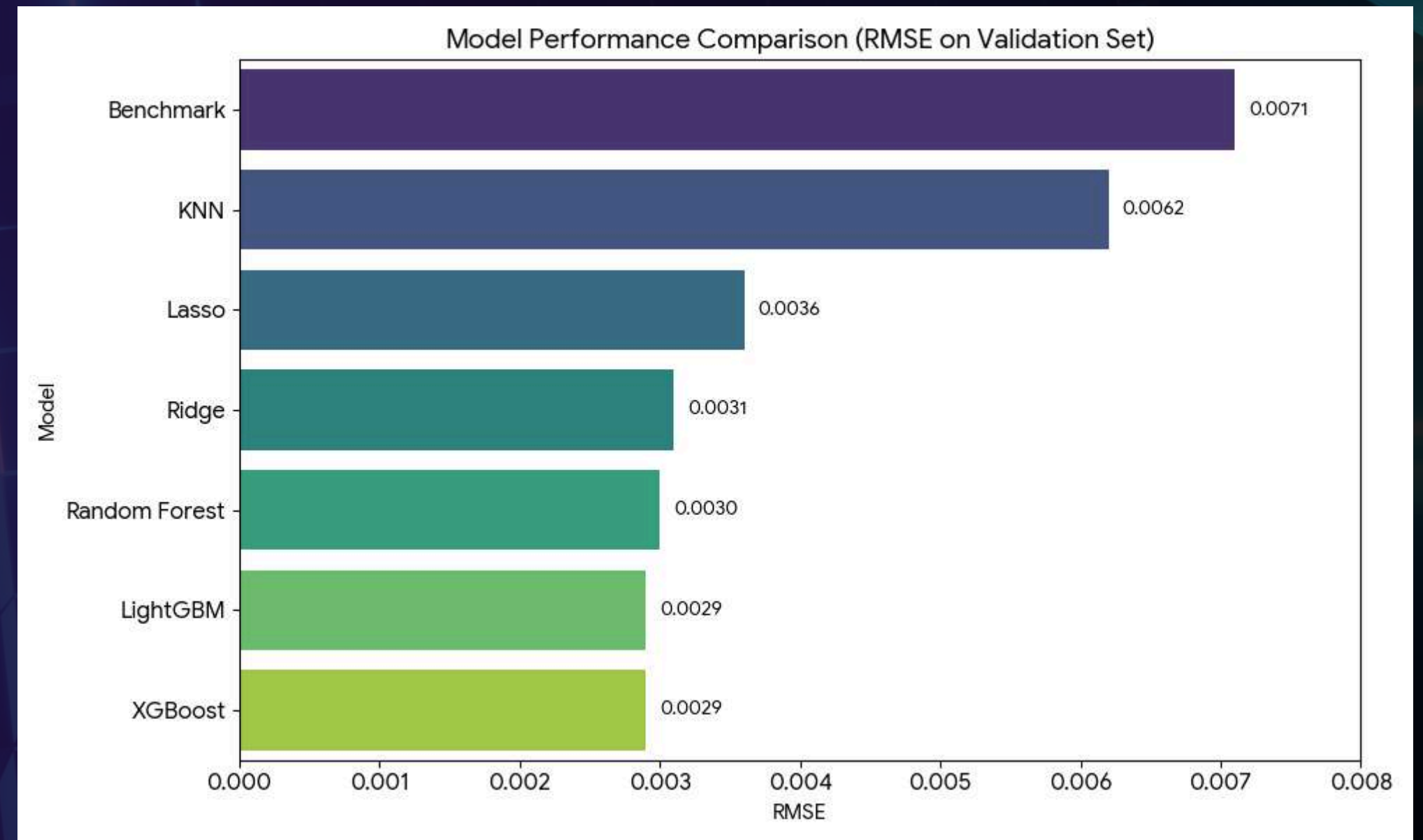
CHAMPION MODEL SELECTION

Selected Model: XGBoost

Performance:

- *RMSE: 0.0029*
- *R^2 Score: 0.95*

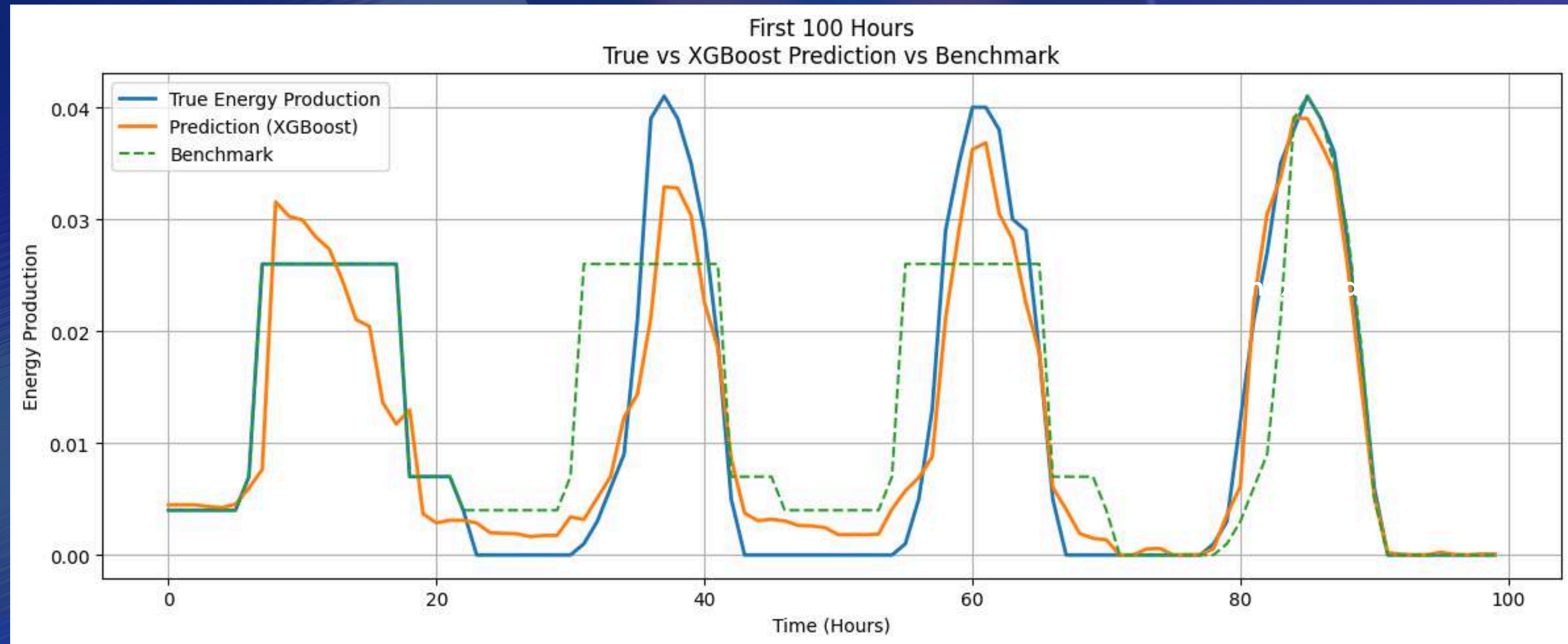
→ Achieved a 59.14% error reduction compared to the Benchmark



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VISUAL VALIDATION

This plot shows a zoom on the first 100 hours of the validation set to inspect model behavior



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STRATEGY GLOBAL MODEL

We use the full dataset to train several machine learning models, such as KNN, Random Forest, and XGBoost, and we select the best-performing model based.

The best machine learning model is xgboost and gives us:

GLOBAL MODEL

RMSE: 0.0029 | R2: 0.9496



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STRATEGY LOCAL MODEL

Plant 0 – Results of models :

```
→ Ridge      | RMSE: 0.00246 | R²: 0.93706
→ Lasso      | RMSE: 0.00293 | R²: 0.91112
→ KNN        | RMSE: 0.00463 | R²: 0.77761
→ RandomForest | RMSE: 0.00244 | R²: 0.93819
→ XGBoost    | RMSE: 0.00243 | R²: 0.93886
/usr/local/lib/python3.12/dist-packages/sklearn
warnings.warn(
→ LightGBM   | RMSE: 0.00243 | R²: 0.93865
```

Final summary for the plants :

	Plant_ID	Best_Model	Best_RMSE	Best_R2
0	0	XGBoost	0.002428	0.938855
1	1	XGBoost	0.002877	0.942469
2	2	LightGBM	0.004795	0.926385
3	3	LightGBM	0.002887	0.946205
4	4	XGBoost	0.001575	0.943771
5	5	XGBoost	0.002545	0.947739
6	6	XGBoost	0.001834	0.944775
7	7	XGBoost	0.001731	0.945938
8	8	XGBoost	0.003276	0.949955
9	9	RandomForest	0.003999	0.943251
10	10	XGBoost	0.003819	0.943376
11	11	XGBoost	0.003889	0.945219
12	12	RandomForest	0.003375	0.945892
13	13	XGBoost	0.003574	0.946190
14	14	XGBoost	0.003393	0.945178
15	15	RandomForest	0.002334	0.942892

Global average of the scores (best models) :

Best_RMSE	0.003021
Best_R2	0.943631



STRATEGY CONCLUSION

Comparison globale vs locale :

	Approach	RMSE	R2
0	Global Model	0.002887	0.949600
1	Local Models (Average)	0.003021	0.943631



4- MODEL INTERPRETATION (XAI)

- *Understanding how the model makes predictions*
- *Identifying the main drivers of solar energy production*
- *Ensuring transparency and trust in the final XGBoost model*



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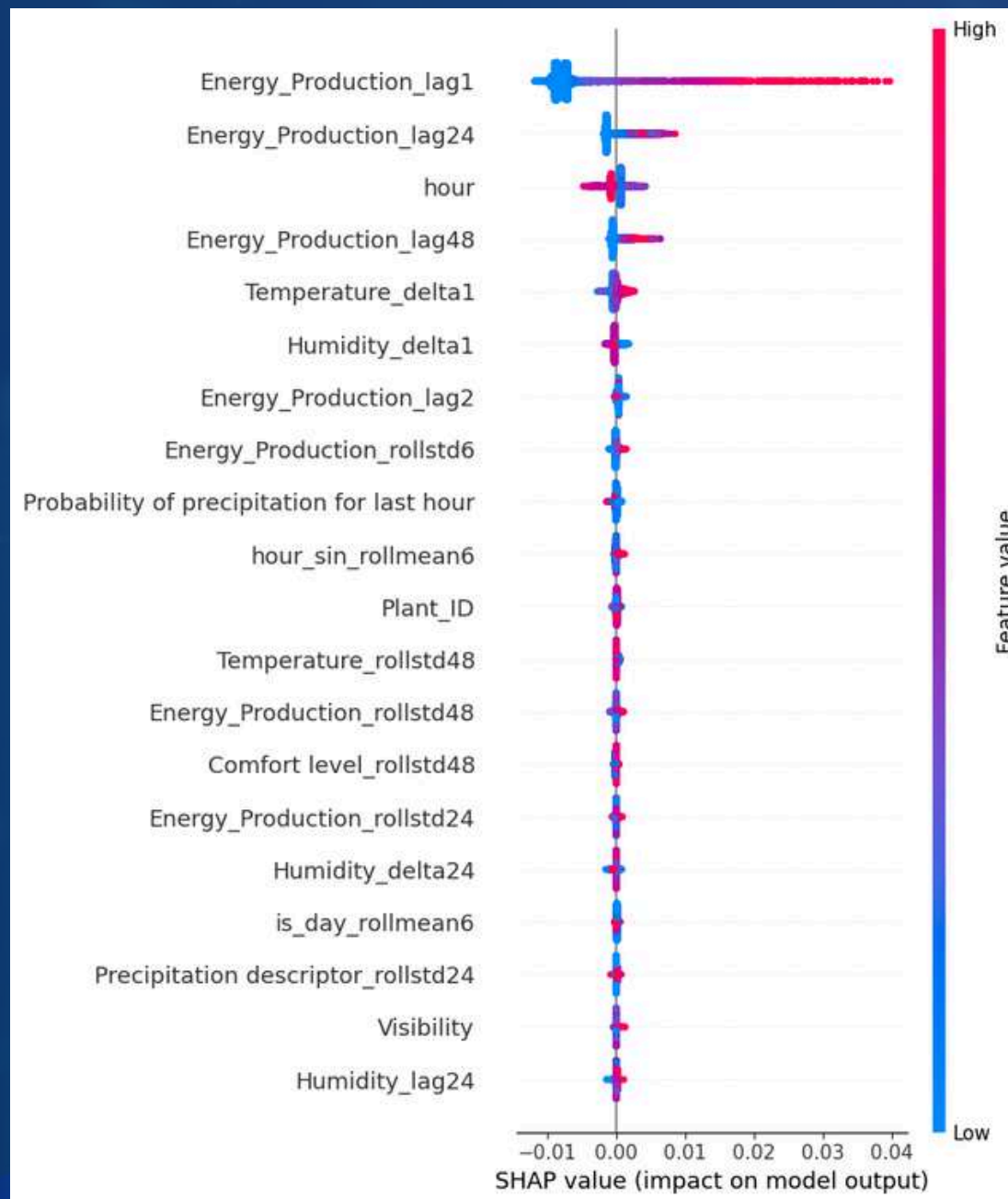
GLOBAL INTERPRETATION USING PERMUTATION IMPORTANCE

- *Model-agnostic interpretability method*
- *Measures performance degradation (RMSE) after feature shuffling*
- *Identifies the most influential drivers of predictions: Recent energy production levels and time-of-day are the dominant drivers of solar energy forecasting, while short-term weather variations and plant-specific effects play a secondary role.*

Rank	Feature	Importance (Mean)
1	Energy_Production_lag1	0.0113
2	hour	0.00108
3	Energy_Production_lag24	0.00089
4	Energy_Production_lag48	0.00087
5	Energy_Production_lag2	0.00046
6	Temperature_delta1	0.0003
7	Humidity_delta1	0.00012
8	Energy_Production_rollstd6	0.00006
9	Energy_Production_rollstd48	0.00004
10	Plant_ID	0.00004
11	hour_sin_rollmean6	0.00004
12	Rainfall amount for last hour_lag1	0.00003
13	Probability of precipitation for last hour	0.00003
14	Energy_Production_rollstd24	0.00002
15	Humidity_rollstd48	0.00002
16	Temperature_rollstd48	0.00002
17	is_day_rollmean6	0.00002
18	Precipitation descriptor_rollstd24	0.00002
19	Energy_Production_rollmean6	0.00001
20	Comfort level_rollstd6	0.00001



GLOBAL MODEL INTERPRETATION (SHAP)



Goal:

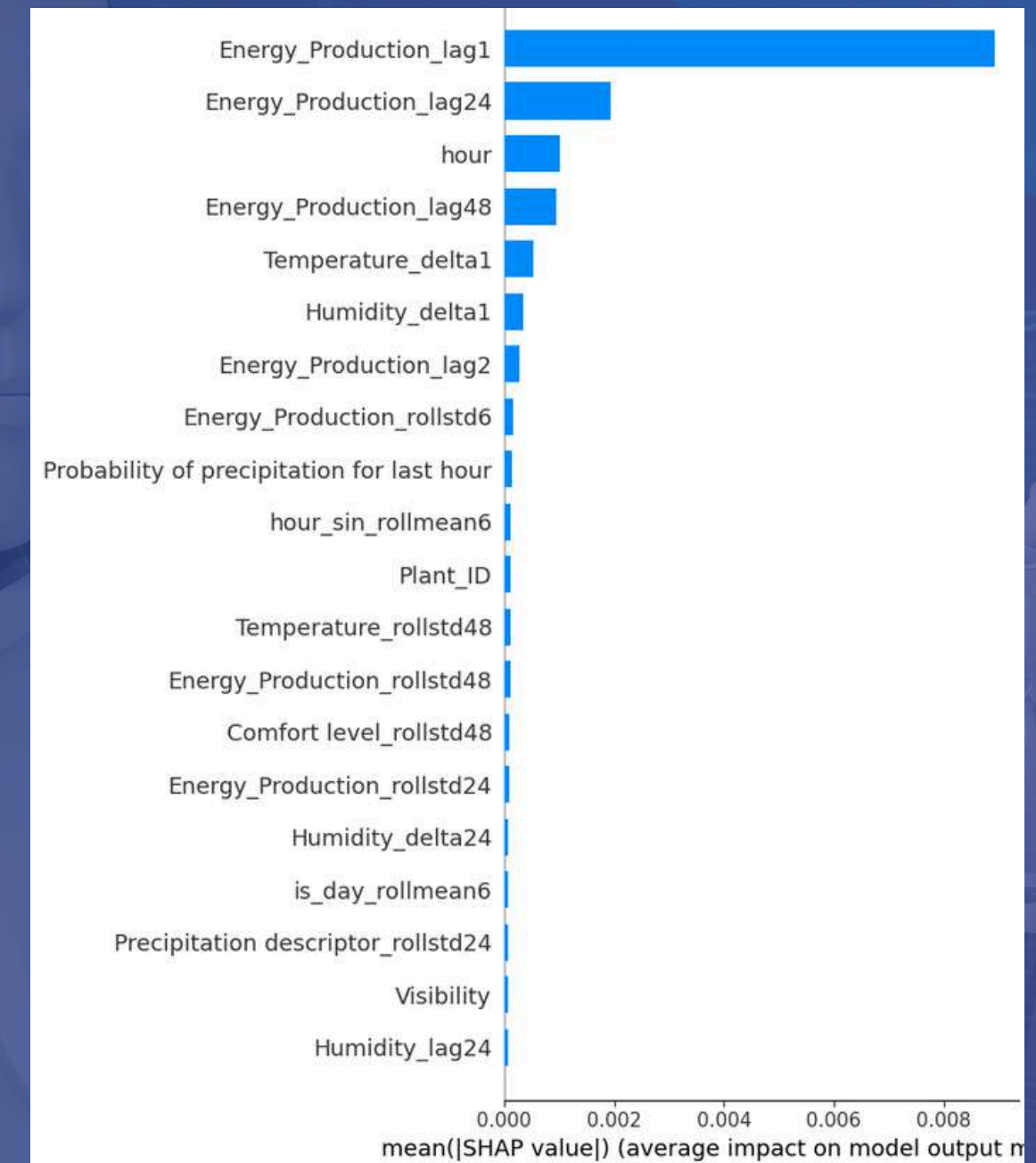
Understand why the model produced a particular prediction (peak production or anomalous error).

Method

- SHAP local explanation on a single observation
- Decomposition of the prediction into feature contributions

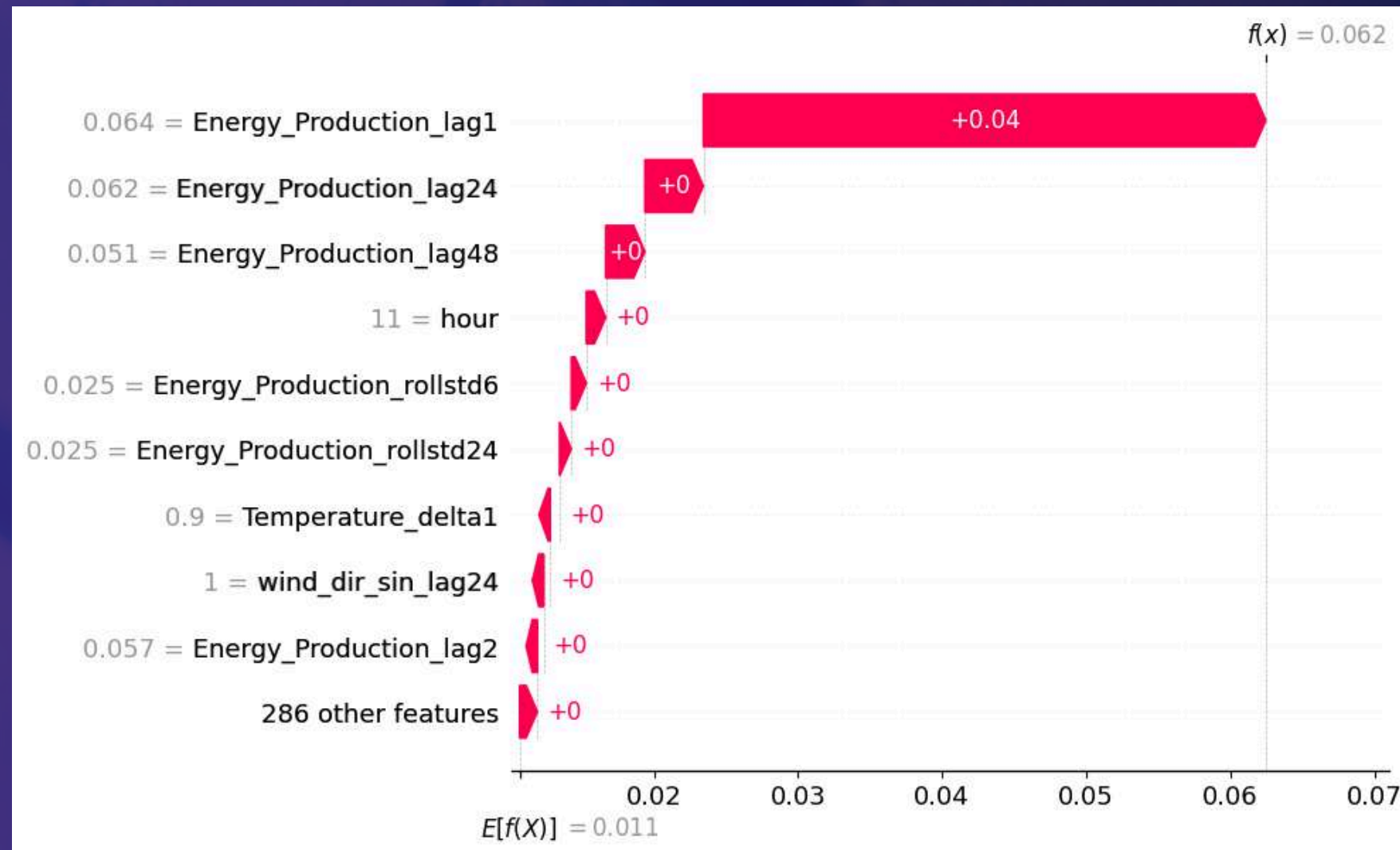
Key Drivers Identified

- Lagged energy production (lag 1, lag 24, lag 48)
- Time of day (hour) reflecting the solar cycle



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LOCAL INTERPRETATION USING SHAP



What does this plot show?

- Explains one specific prediction of the XGBoost model
- Starts from the average prediction:
- $E[f(X)] = 0.011$
- Each bar shows how a feature pushes the prediction up or down
- Final prediction: $f(x) \approx 0.062$

Key drivers:

- Energy_Production_lag1 (+0.04)
- → Strong short-term persistence (previous hour drives current output)
- Energy_Production_lag24
- → Daily seasonality in solar production (24h cycle)



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LOCAL ANOMALY INTERPRETATION

What does this plot show?

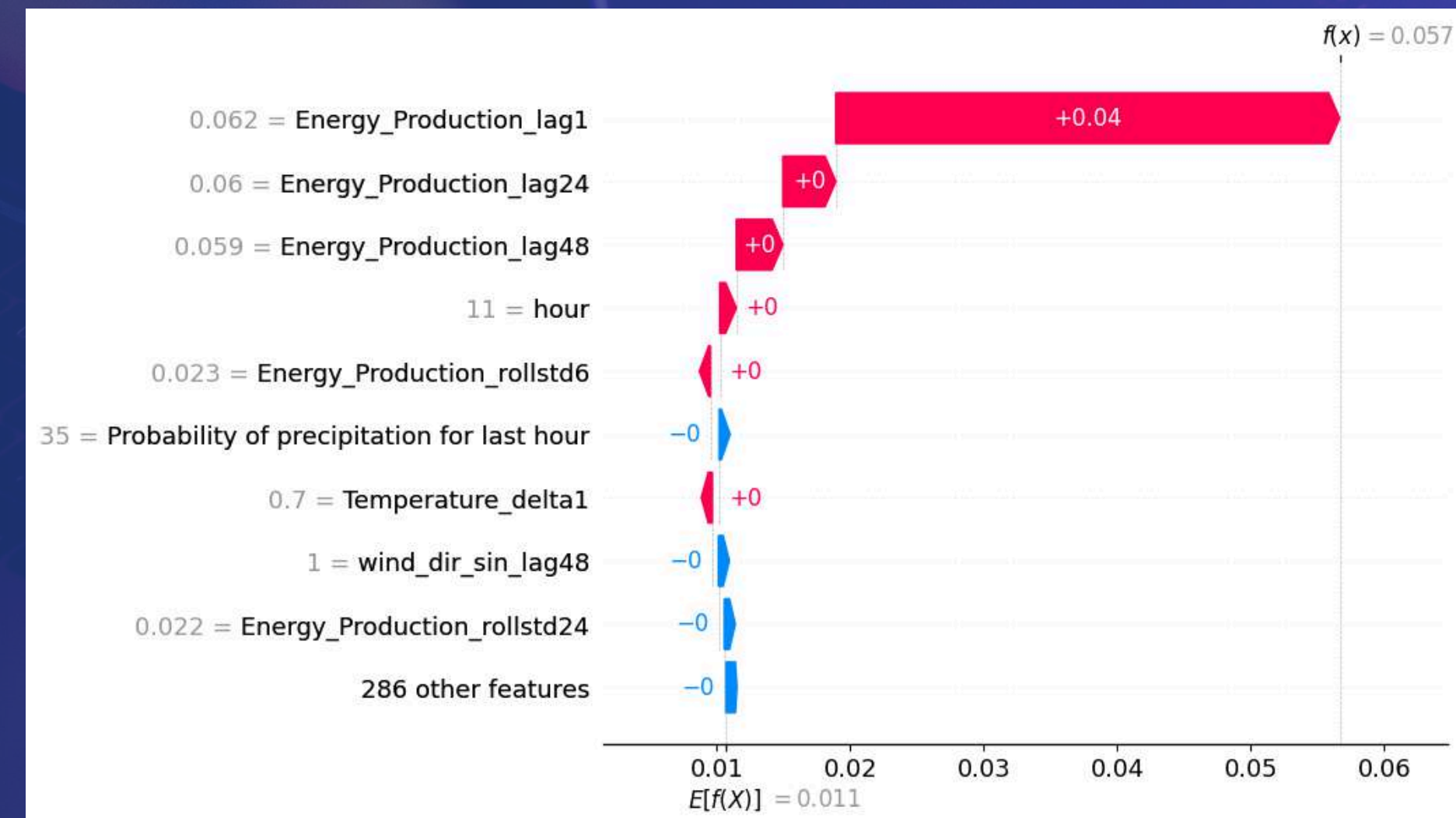
- Explains the most anomalous prediction
- Selected as the observation with the largest absolute error

Even in anomalous cases, the model mainly relies on

- recent historical production

Prediction errors are likely linked to

- atypical meteorological or operational conditions
- The model remains consistent and interpretable



5- CONCLUSION

Model	RMSE	R ²
Benchmark	0.0064	0.7506
XGBoost	0.0029	0.9496

RMSE Improvement vs Benchmark: 55.04% | Final model: XGBoost

Final Verdict

- ✓ 55% RMSE improvement
- ✓ R² close to 0.95
- ✓ XGBoost selected

Business Value

- Better short-term planning
- Improved grid stability
- Early anomaly detection

Next Steps

- Add physical drivers
- Strengthen robustness
- Plant-level optimization



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**Thank You for
your attention !**